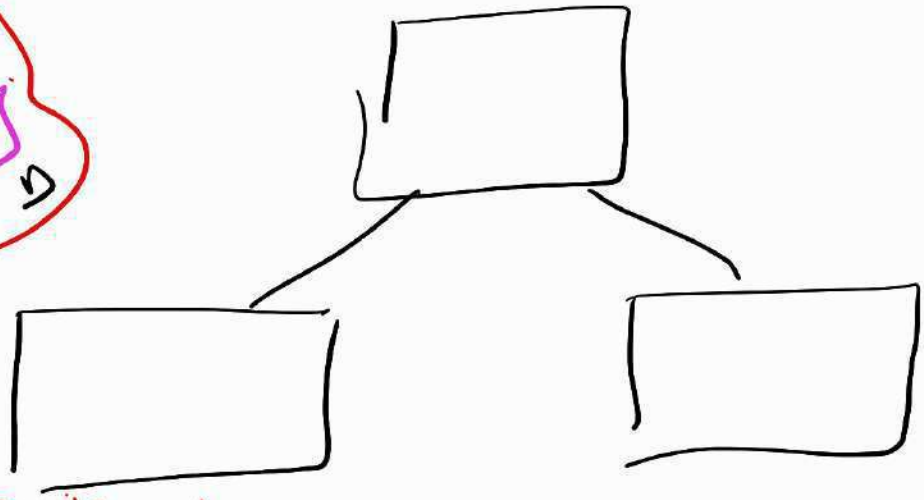
Informed Search:-

→ Utilizes additional knowledge to guide search process while the other explores without specific information (or) guidance.

State-Space Search:-

Given a State Space.



Search Algorithm:-

How to intelligently explore a state space by following state transformation rules.

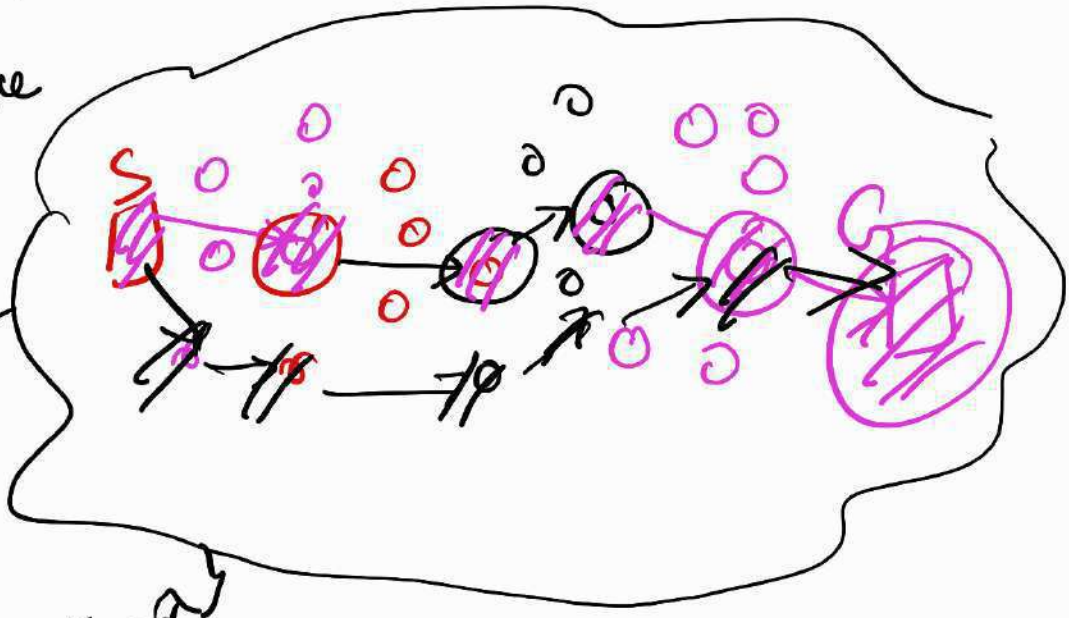
→ Choose an suitable search algorithm from the Class of search problems.

→ Size of State Space.

Domain-Independent Algorithms

User needs to understand the domain description and adopt to suitable problem solving.

State-Space Search



MoveGen(S) – A domain function

Goaltest(N) – A domain function

MoveGen(N) → {_, _, _}

Define a State :-

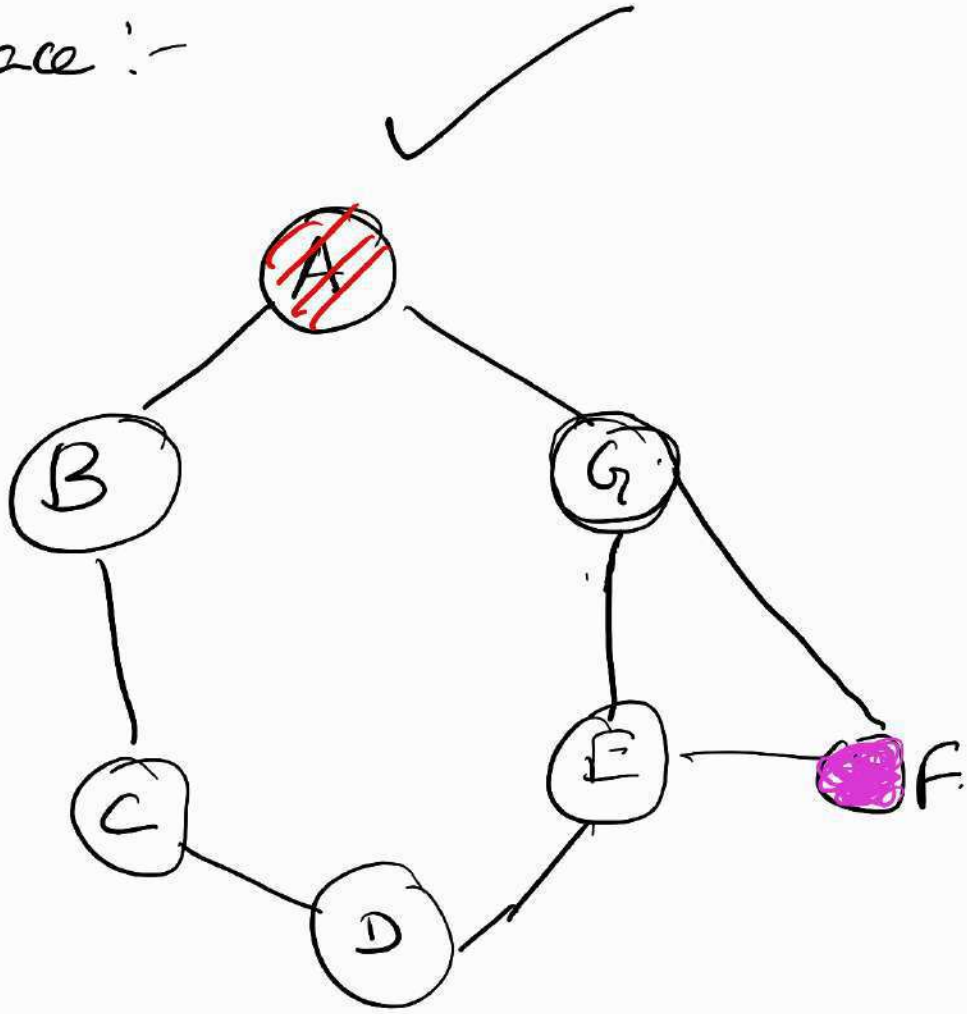
A Search algorithm must choose a from set of Candidates.

OPEN:-

Search Algorithm:-

1. Generate & Test
2. Traverse.
3. Check.

State Space:-



Search Space:-

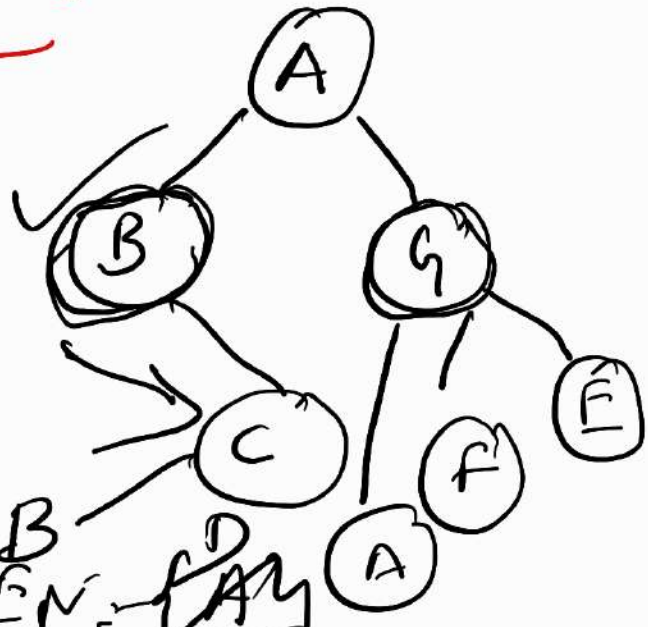
S :- { A }

G :- { F }

MoveGen( )

A	{B, G}
B	{A, C}
C	{B, D}
D	{C, E}
E	{F, G, D}
G	{A, F, E}
F	{E, G}

Search Tree:-



OPEN :- ~~{A}~~

OPEN = { B, G }

OPEN = { G, C, D }

Closed :- { A, B }

basic search( ) // with Closed set

OPEN ← {S}
Closed ← { }
While OPEN is not empty.

(Head, tail) do pick some 'N' from OPEN

UPEN ← OPEN − {N}

Closed ← Closed ∪ {N}

IF Goal Test(N) = TRUE

then return N

else

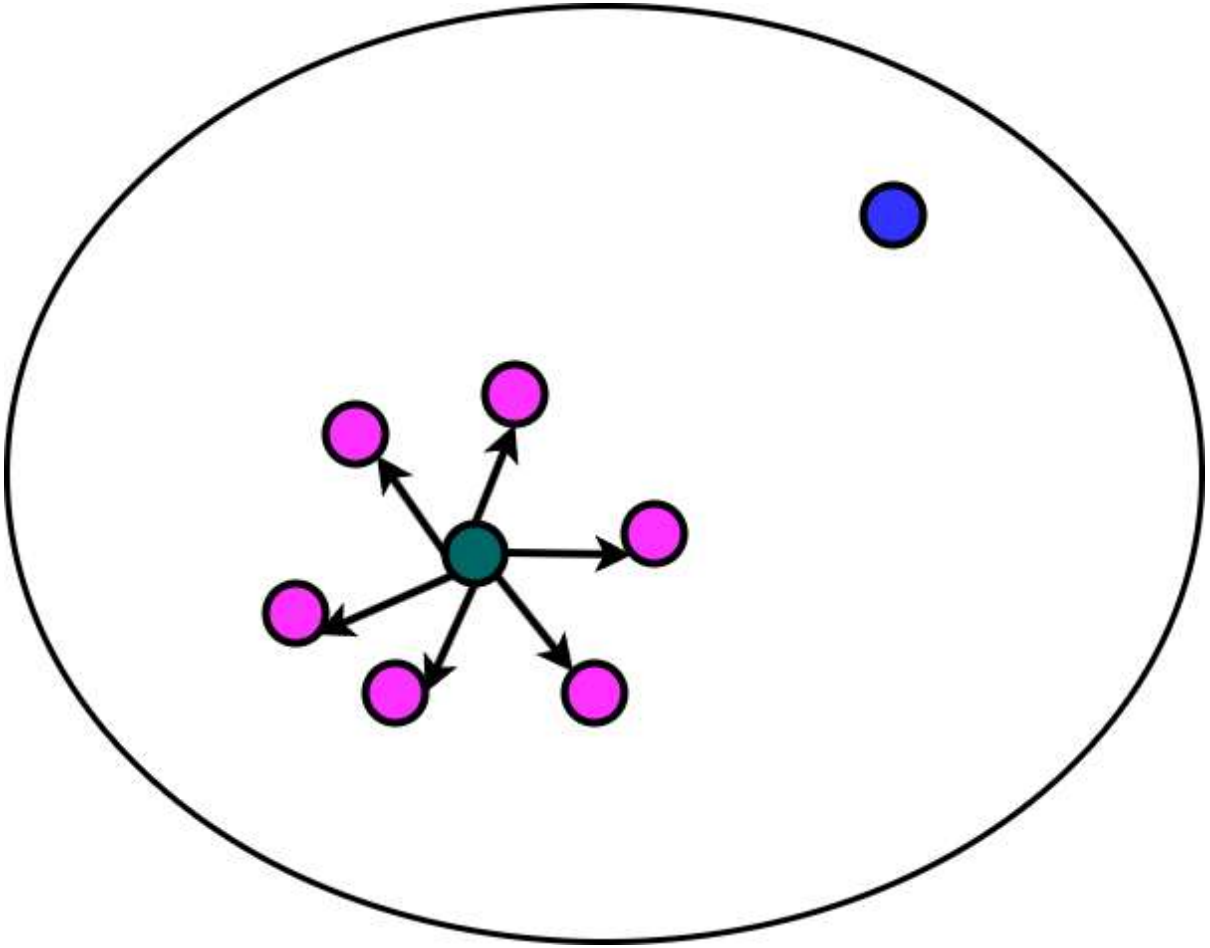
OPEN ← OPEN ∪ {MOVEGEN(N) − Closed}

SET DIFFERENCE

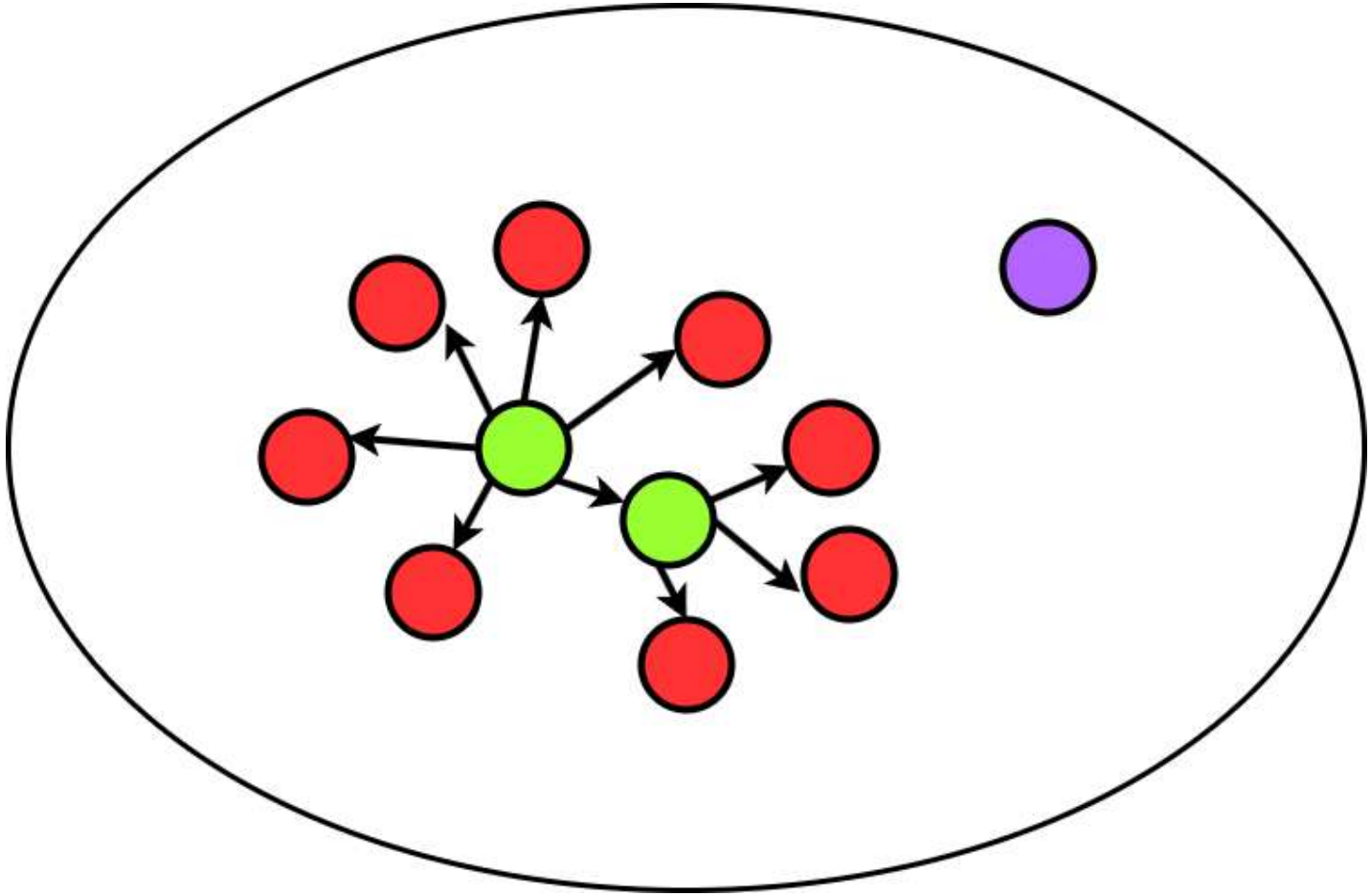
return failure.

+ OPEN( )

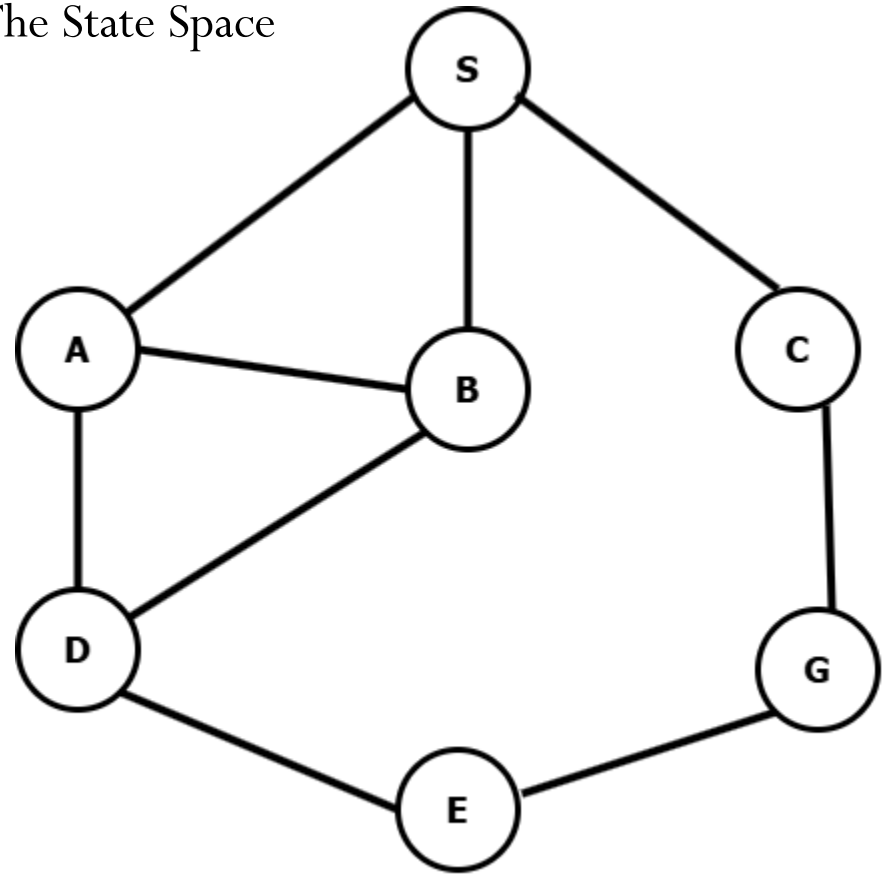
Moves:State transformation



The set OPEN of candidates



The State Space



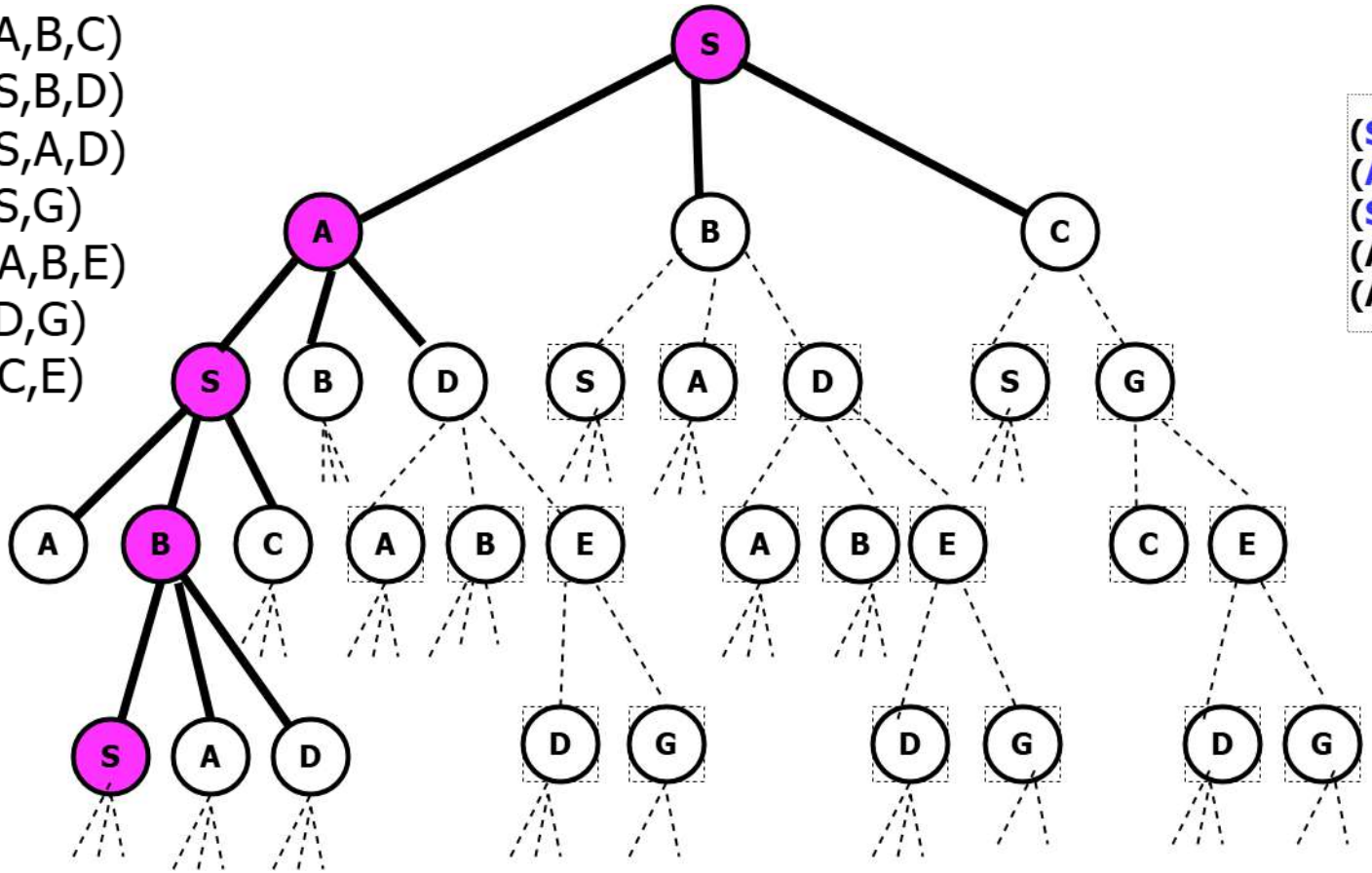
The MoveGen Function

S->(A,B,C)
A->(S,B,D)
B->(S,A,D)
C->(S,G)
D->(A,B,E)
E->(D,G)
G->(C,E)

Basicsearch

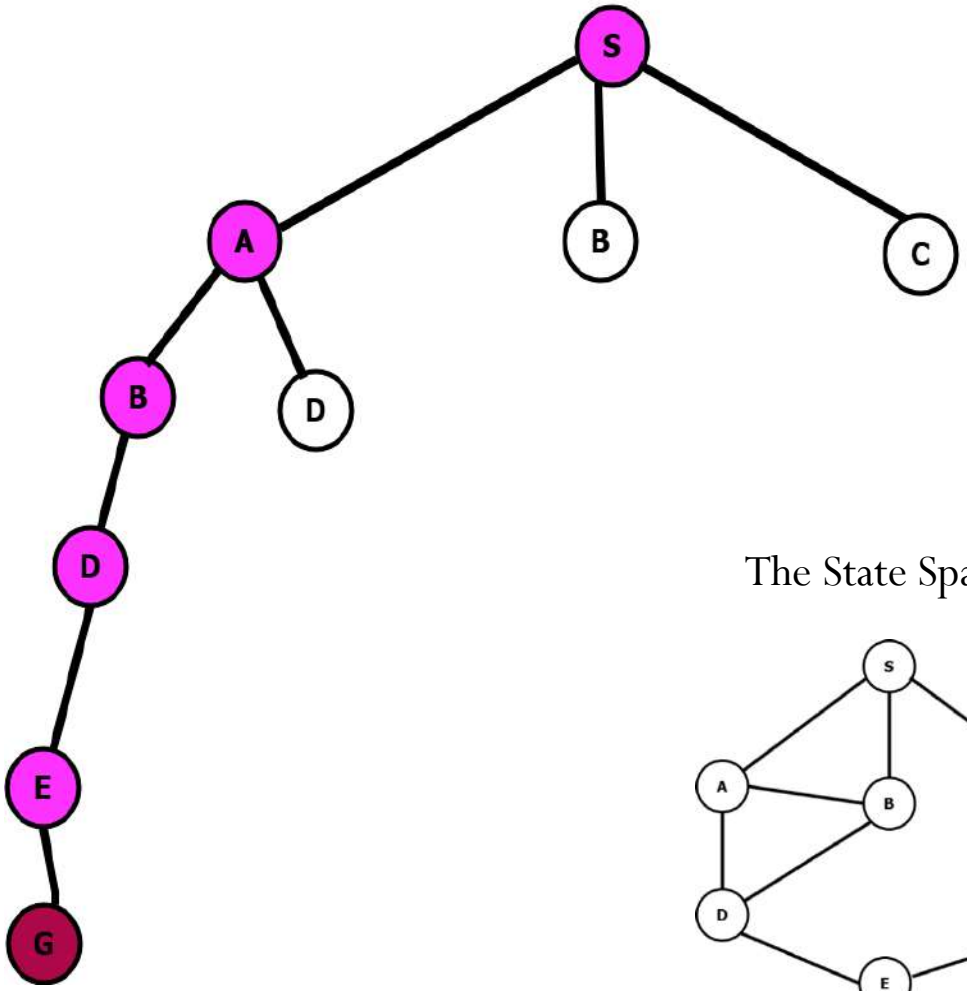
The MoveGen Function

S->(A,B,C)
A->(S,B,D)
B->(S,A,D)
C->(S,G)
D->(A,B,E)
E->(D,G)
G->(C,E)



(S)
(ABC)
(SBDBC)
(ABCBDBC)
(ASADCBDBC)

Search Tree for a basic search2

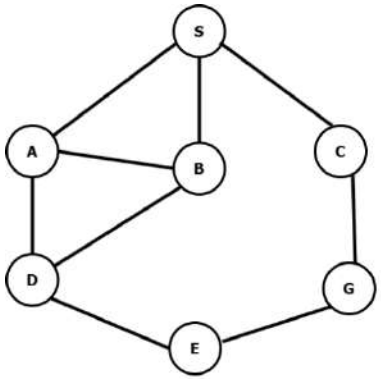


OPEN	CLOSED
(S)	()
(ABC)	(S)
(BDBC)	(AS)
(DDBC)	(BAS)
(EDBC)	(DBAS)
(GDBC)	(EDBAS)

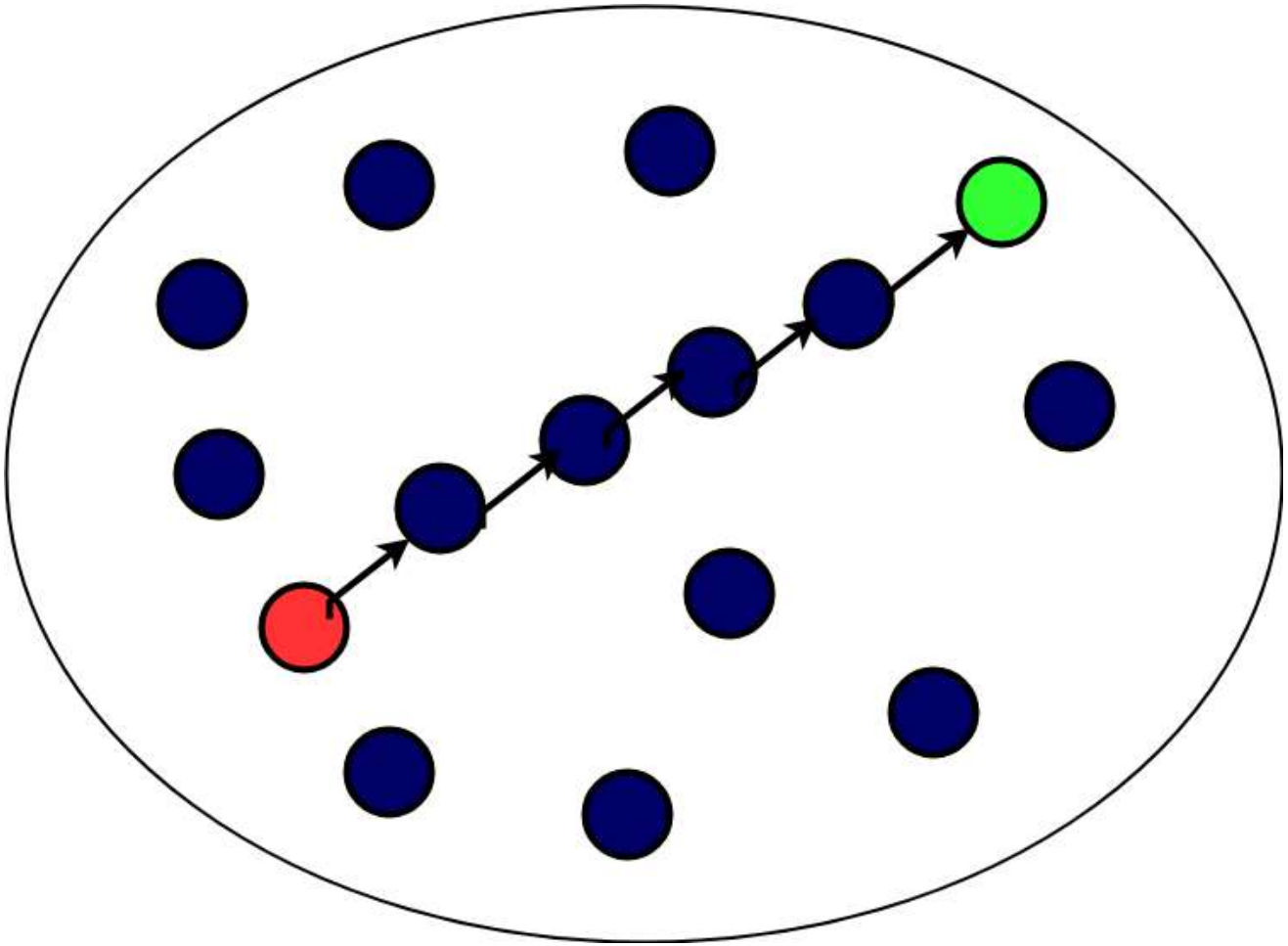
The MoveGen Function

S->(A,B,C)
A->(S,B,D)
B->(S,A,D)
C->(S,G)
D->(A,B,E)
E->(D,G)
G->(C,E)

The State Space



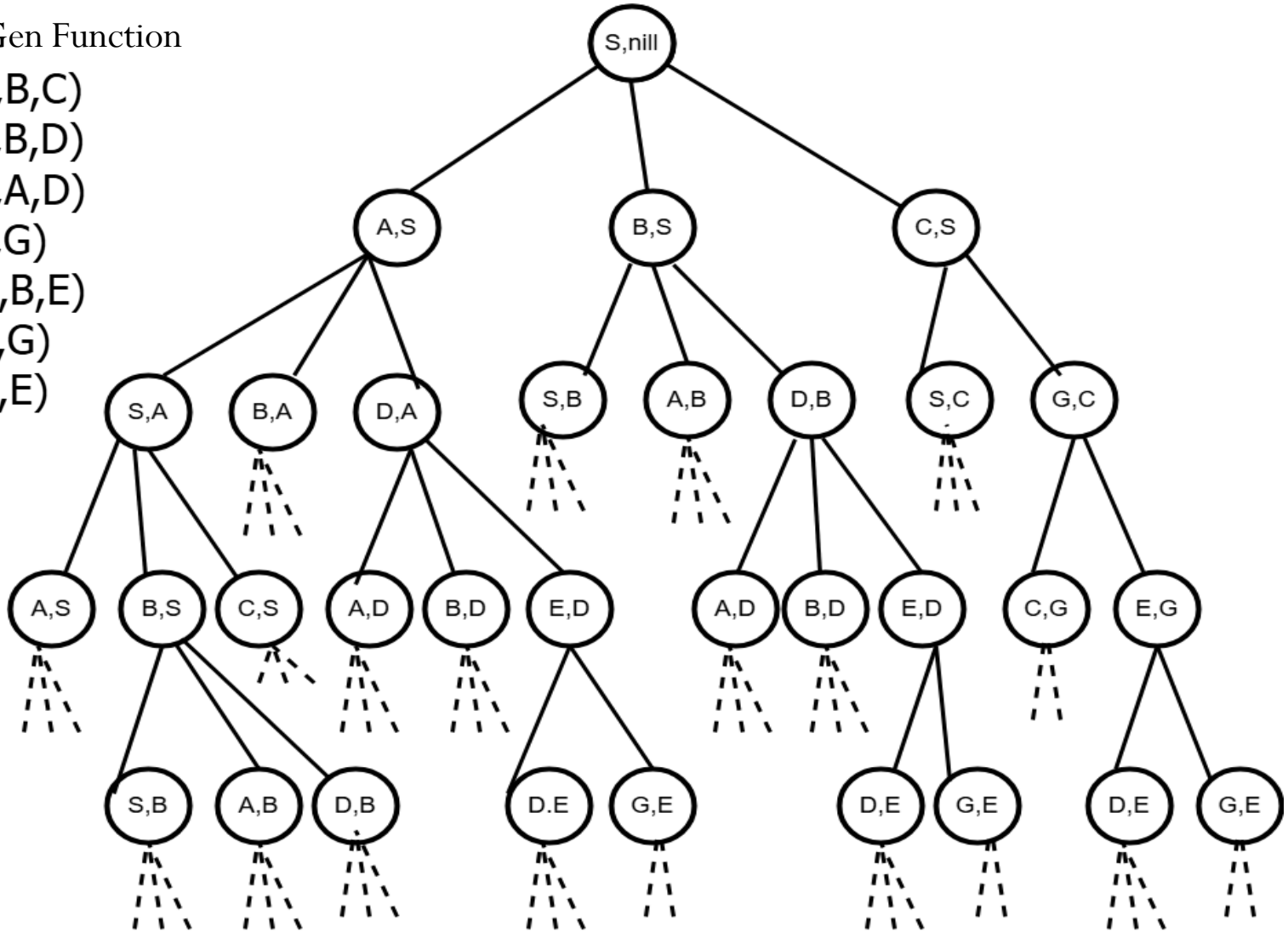
A Solution



Node Pairs in the Search Tree

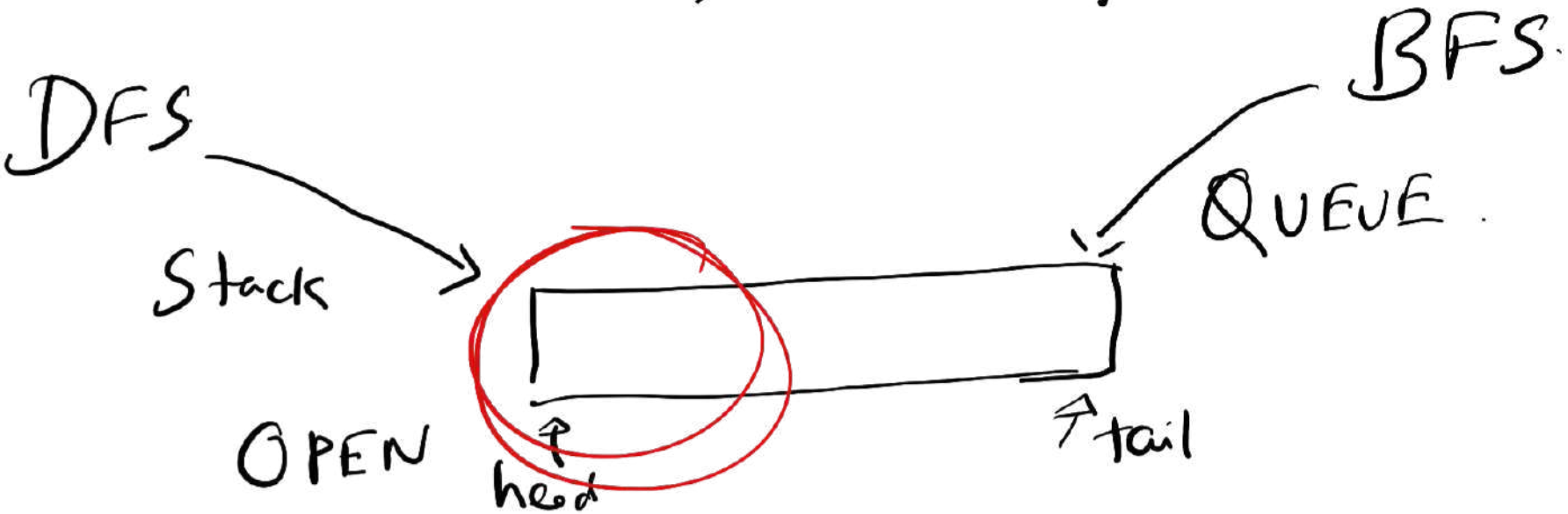
The MoveGen Function

S->(A,B,C)
A->(S,B,D)
B->(S,A,D)
C->(S,G)
D->(A,B,E)
E->(D,G)
G->(C,E)

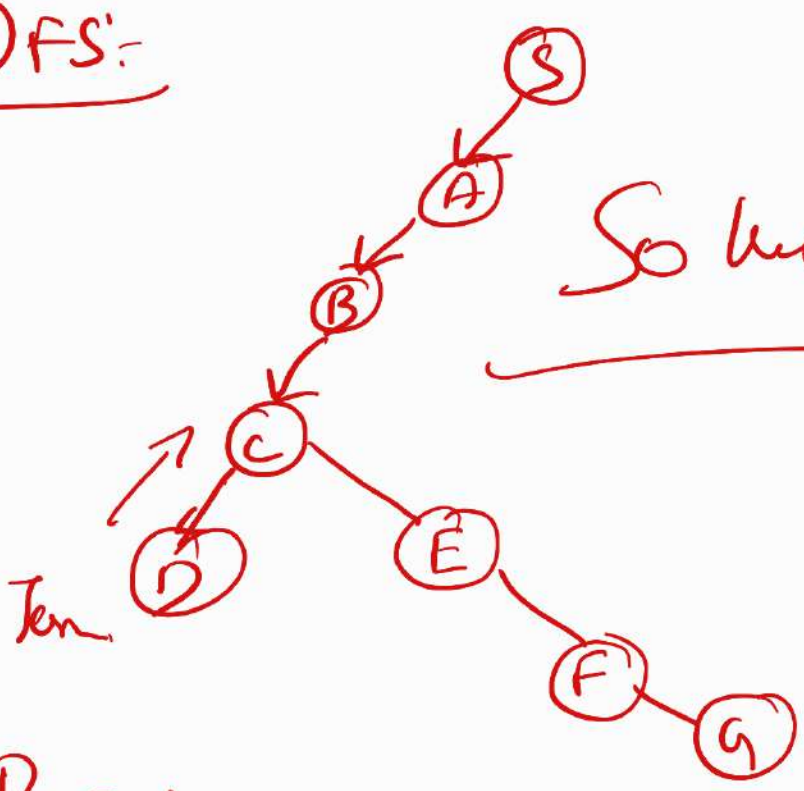


In BFS & DFS:-

* Instead of set, linked list datastructure is used
* So, the exploration of state starts from head node.
* Both BFS & DFS explore the same search tree. But, order is different.



DFS:-

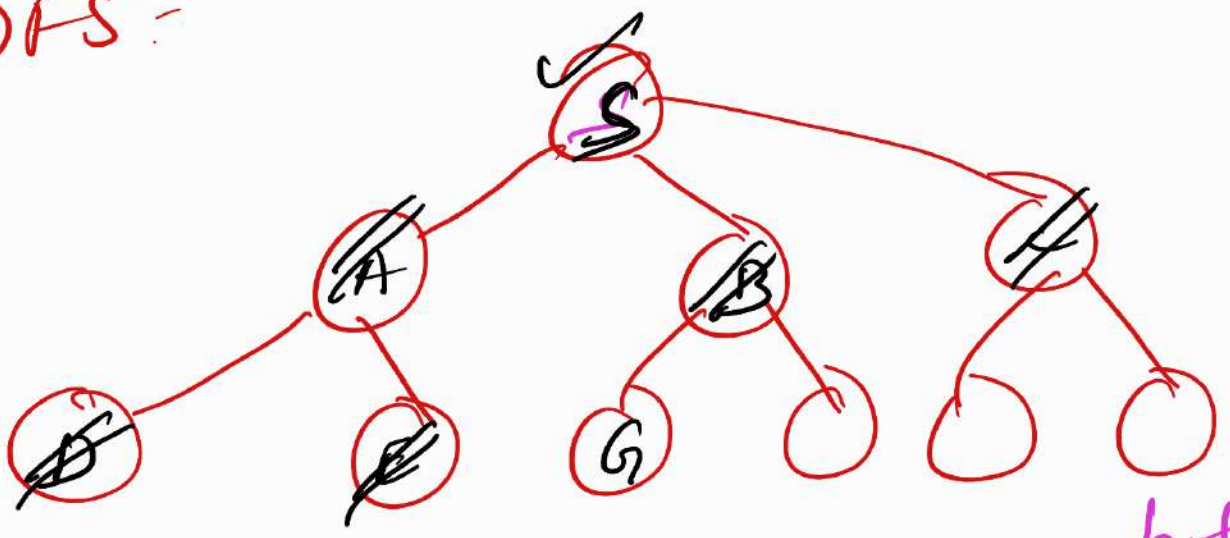


G.

Solution path

OPEN:-

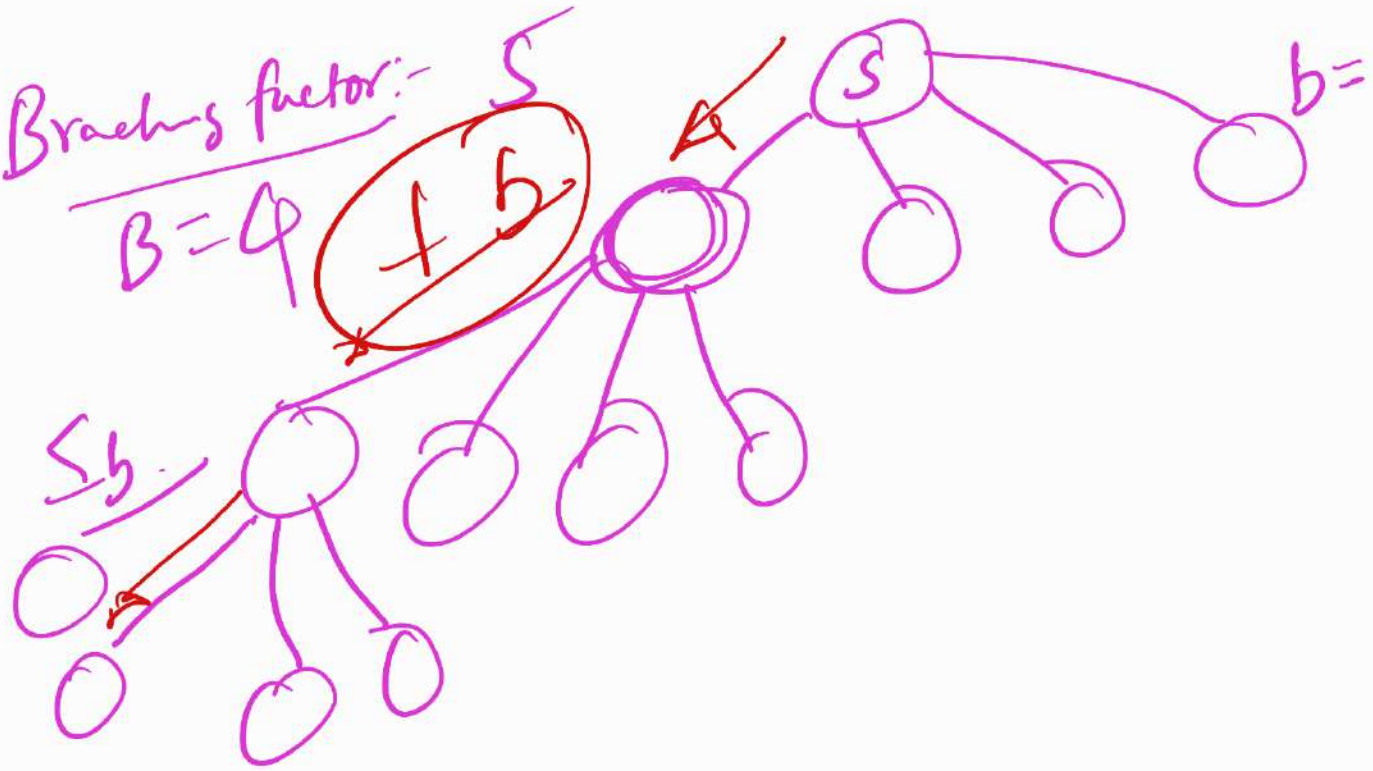
BFS:-



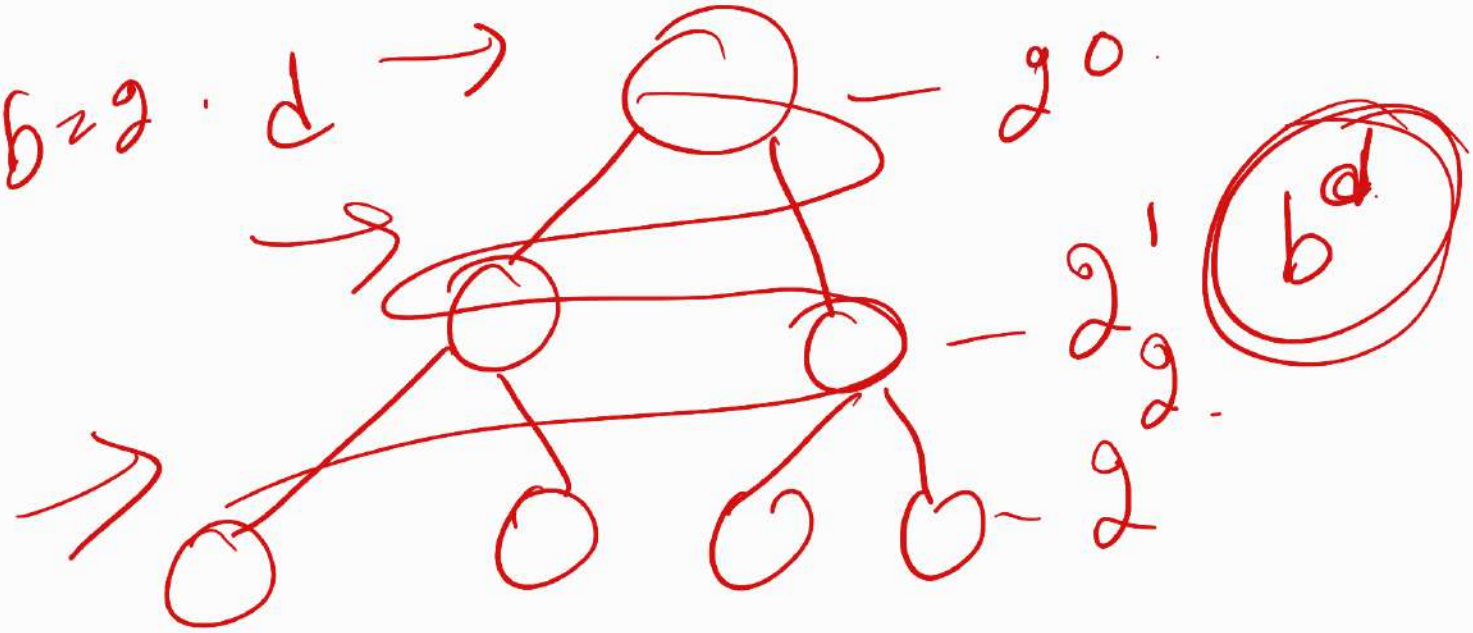
b+b+b

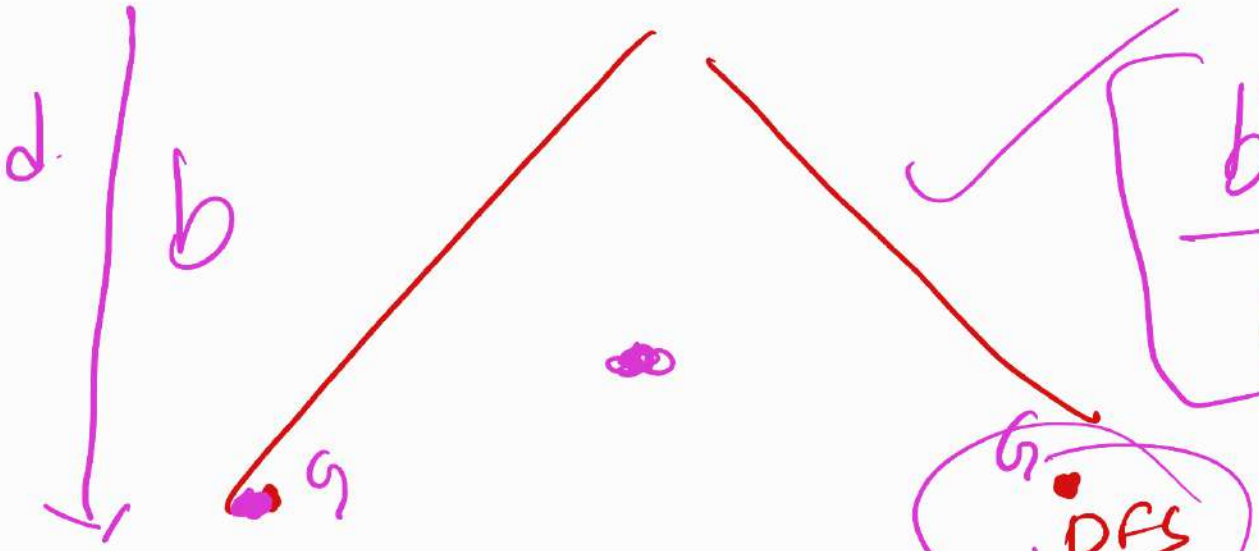
Branching factor:-

B=4



$b = 2 \cdot d \rightarrow$





$$\left[\frac{b^d - 1}{b - 1}\right]$$

DFS

BFS. Best

Worst DFS

BFS.

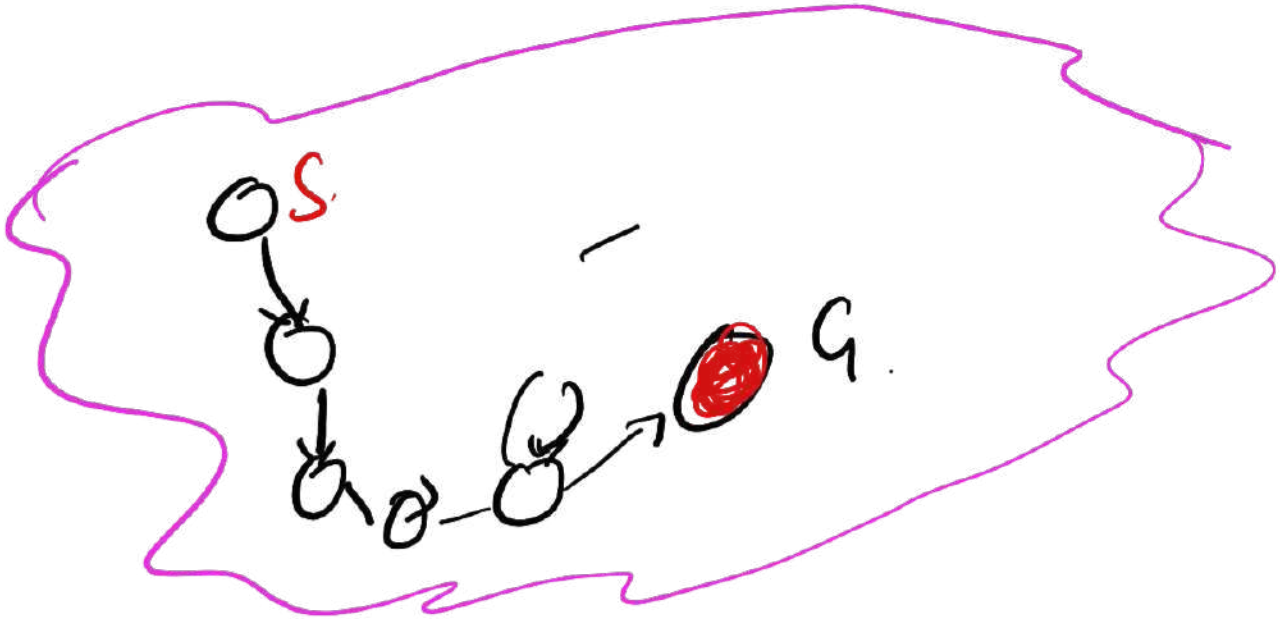
$$\delta_{DFS} = d + 1 \quad \Omega_{DFS} = \left[\frac{b^{d+1} - 1}{b - 1}\right]$$

$$\delta_{BFS} = \left[\frac{b^d - 1}{b - 1}\right] + 1 \quad \Omega_{BFS} = \left[\frac{b^{d+1} - 1}{b - 1}\right]$$

Exponentially $\rightarrow$ $O(b^d)$ $O(b^d)$

Convergence

DFS:-



BFS:-

DFS(S) OPEN=STACK

```
OPEN ← (S, null) : [ ]
CLOSED ← empty list
while OPEN is not empty
    nodePair ← head OPEN
    (N, _) ← nodePair
    if GoalTest(N) = TRUE
        return ReconstructPath(nodePair, CLOSED)
    else CLOSED ← nodePair : CLOSED
        children ← MoveGen(N)
        newNodes ← RemoveSeen(children, OPEN, CLOSED)
        newPairs ← MakePairs(newNodes, N)
        OPEN ← newPairs ++ (tail OPEN)
return empty list
```

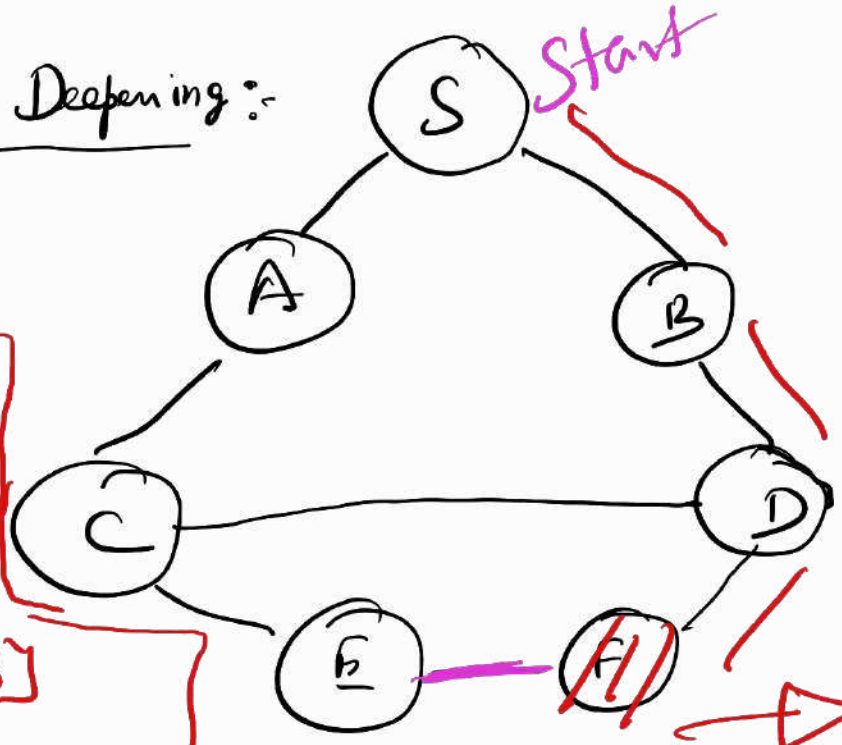
For BFS(S): //OPEN=QUEUE

Replace the line 13 in DFS (S) , with

OPEN<----- (tail OPEN) ++ new Pairs // So, QUEUE datastructure

(Linear/Exponential	DFS	BFS.
TIME	Exponential	Exponential.
SPACE	Linear	Exponential.
Solution	No Guarantee (Infinite Space.)	Shortest Path.
Completeness	Not for Infinite Search Space	Path exists

Iterative- Deepening :-



MONEGEN

S → {A, B}

A → {S, C}

B – {S, D}

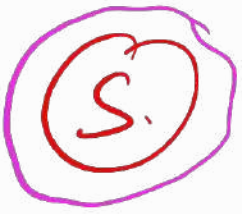
C → {A, D, E}

D → {B, C, F}

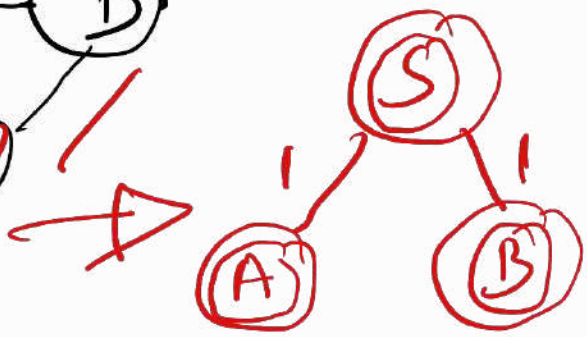
E → {F, C}

F → {E, D}

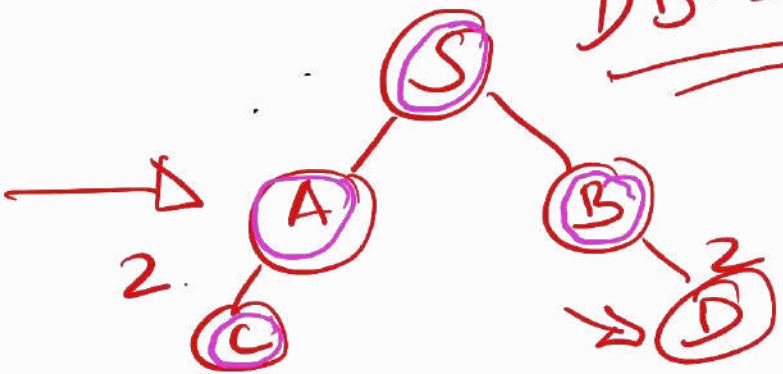
DB-0



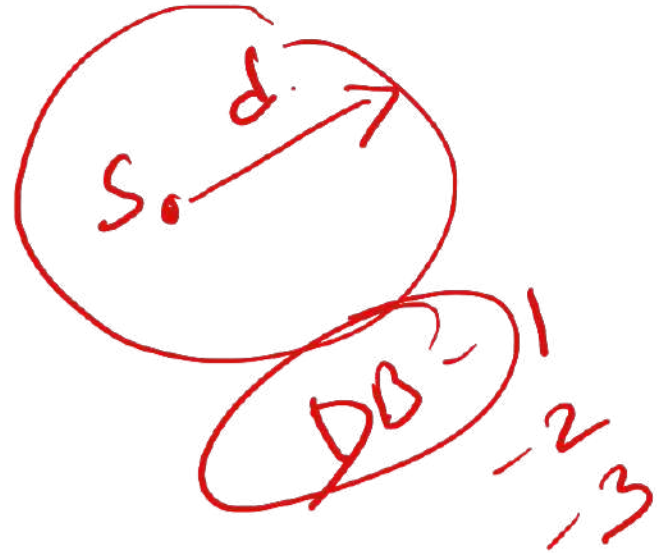
DB-1



DB-2

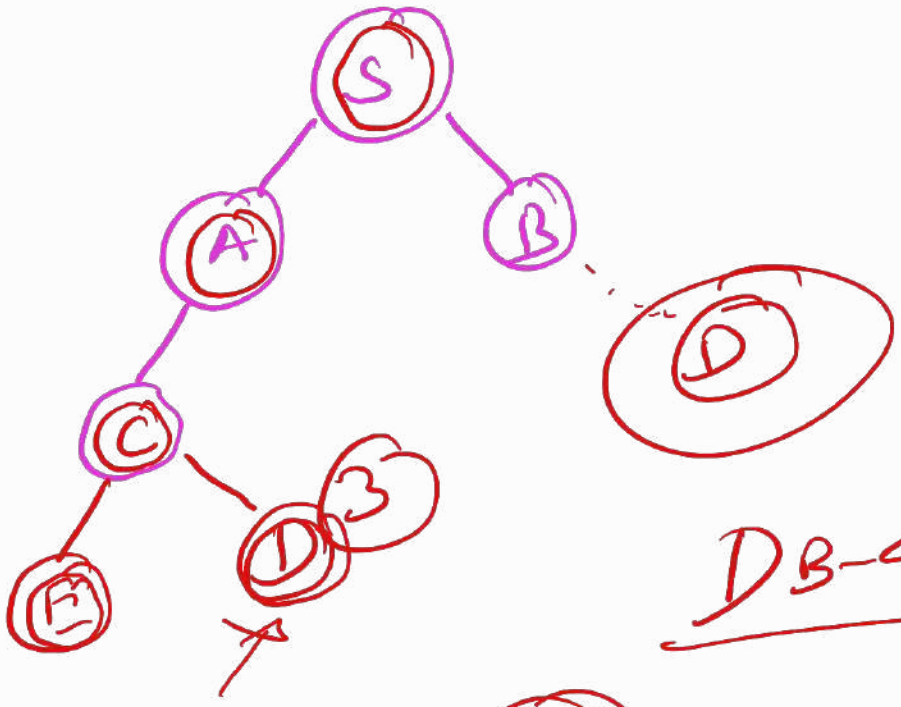


```
DB-DFS(S, depthBound)
OPEN ← (S, null, 0) : [ ]
CLOSED ← empty list
while OPEN is not empty
    nodePair ← head OPEN
    (N, _, depth) ← nodePair
    if GoalTest(N) = TRUE
        return ReconstructPath(nodePair, CLOSED)
    else CLOSED ← nodePair : CLOSED
        if depth < depthBound
            children ← MoveGen(N)
            newNodes ← RemoveSeen(children, OPEN, CLOSED)
            newPairs ← MakePairs(newNodes, N, depth + 1)
            OPEN ← newPairs ++ tail OPEN
        else OPEN ← tail OPEN
return empty list
```

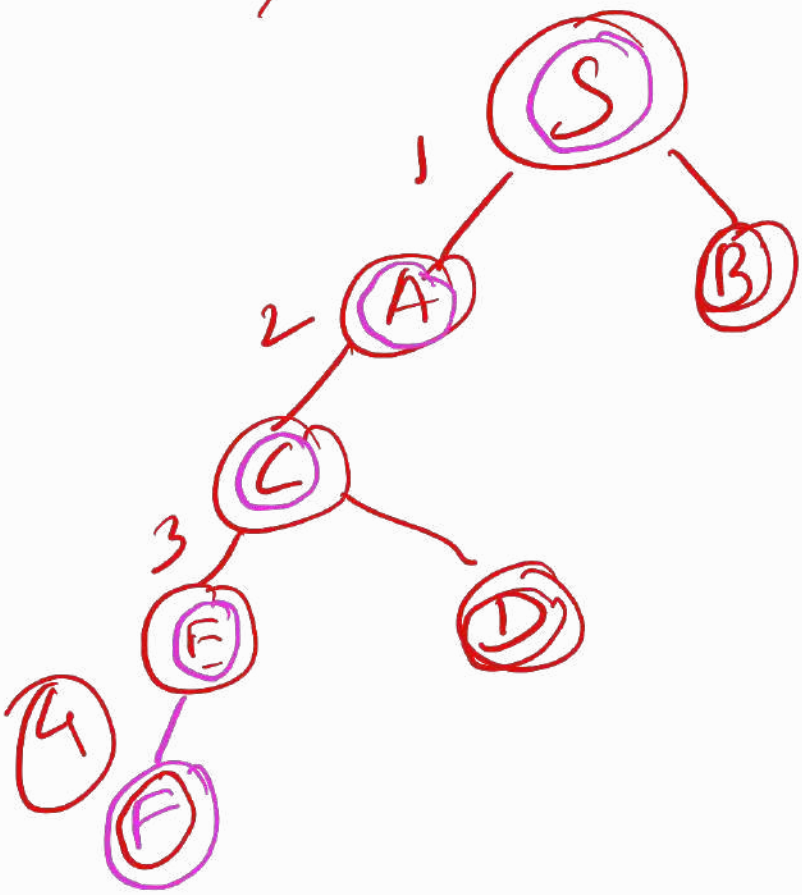


1<1

DB-3



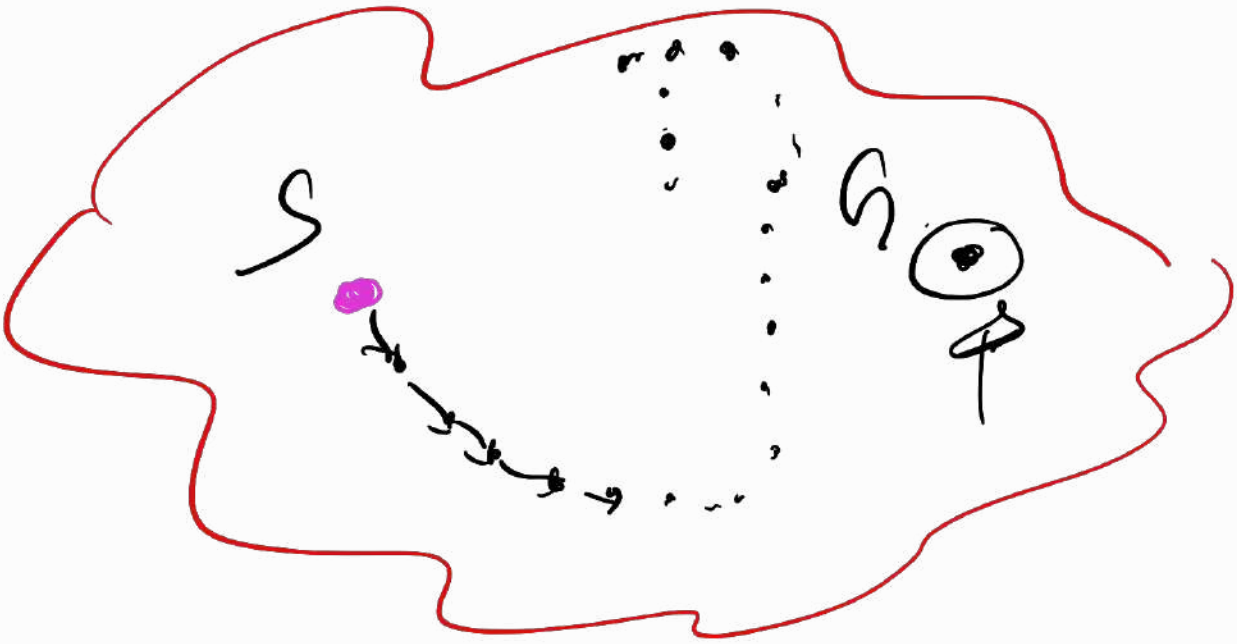
DB-4



```
DFID (Start)
    depthbound ← 1
    While true
        do  DepthboundedDFS (Start, depthbound)
            depthbound ← depthbound + 1
```

* DFID does a ~~Series~~ repetitive cell of DBDFS with ↑ depthbound

Both BFS & DFS are oblivious of the goal.

* Predetermined Trajectory.
* a metric about distance is required.

Informed Strategies